$$x = x_0 + v_0 + \frac{1}{2}at^2$$
$$v = v_0 + at$$
$$v^2 = v_0^2 + 2a(x - x_0)$$
$$\vec{v} \overset{\text{def}}{=} \frac{d\vec{r}}{dt}, \vec{a} \overset{\text{def}}{=} \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$
$$a_p = \frac{v^2}{r} = \omega^2 r, a_t = \frac{dv}{dt} = \frac{d\omega}{dt}r$$
$$P = \frac{dW}{dt} = Vi = \tau\omega, P_{ave} = \frac{\Delta W}{\Delta t} = \vec{F} \cdot \vec{V}_{ave}$$
$$\left|\vec{F}_f\right| = \mu|\vec{N}|$$
$$\left|\vec{D}\right| = \frac{1}{2}C\rho A v^2$$
$$\vec{F}_s = -k\Delta\vec{x}$$

$$\left|\vec{F}\right| = \frac{GM_1M_2}{r^2}$$
$$\vec{F}_{Total} = \frac{d\vec{p}}{dt}$$
$$\vec{\tau} = \vec{r} \times \vec{F} = |\vec{r}|\left|\vec{F}\right|Sin(\theta), \vec{\tau}_{net} = I\vec{\alpha}$$
$$W_{Total} = \Delta KE$$
$$\Delta KE = W_{cons} + W_{non-Cons}$$
$$\Delta E_{mach} = \Delta KE + \Delta PE = W_{non-Cons}$$

$$W = \Delta E_{mach} + \Delta E_{Th} + \Delta E_{int}$$
$$\vec{p} = m\vec{v}, \qquad \vec{p}_{Total} = \sum_i \vec{p}_i = M\vec{v}_{CM}$$

$$\Delta\vec{p} = \vec{F}\Delta t = \vec{J} = \int_{t_1}^{t_2} \vec{F}\, dt$$

$$\frac{d\vec{p}}{dt} = \dot{m}\vec{v} + m\vec{a}$$
$$v_{1A} = \frac{m_1 - m_2}{m_1 + m_2}v_{1B} + \frac{2m_2}{m_1 + m_2}v_{2B}$$

A = after

$$R \overset{\text{def}}{=} -\frac{dM_{Rocket}}{dt}(Rocket\ Mass\ loss\ rate)$$

$$T \overset{\text{def}}{=} Rv_{rel}(Thrust(N))$$
$$T = M_{Rocket}a_{Rocket} - First\ Rocket\ Equation$$

$$v_f = v_i + v_{Rel}\ln(\frac{Mi}{Mf}) - Second\ Rocket\ Equation$$

$$\rho = \frac{m}{V}, m = \int \rho dV$$
$$\alpha = \frac{\Delta L}{L\Delta T}, \beta = \frac{\Delta V}{V\Delta T} = 3\alpha, Q = \pm mL$$

$$v = -\frac{Latteral\ Strain}{Longitudinal\ Strain} = -\frac{\frac{\Delta W}{W}}{\frac{\Delta L}{L}}$$
$$Comp.wave\ spd: \alpha = \frac{\sqrt{\lambda + 2\mu}}{\rho}$$
$$Transv.wave\ spd: \beta = \sqrt{\frac{\mu}{\rho}}$$
$$W = \frac{1}{2}AE\frac{\Delta L^2}{L}$$

$$u = \frac{1}{2}G\left(\frac{\Delta x}{L}\right)^2\left(\frac{J}{m^3}\right)(Shear)$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$$
$$\omega = \omega_0 + \alpha t$$
$$\omega^2 = \omega^2 + 2\alpha(\theta - \theta_0)$$
$$|\vec{\omega}| = \frac{d\theta}{dt} = \frac{2\pi}{T} = \frac{v}{r}, |\vec{\alpha}| = \frac{d\omega}{dt} = \frac{a}{r}$$
$$\omega\left[\frac{rad}{s}\right] = rev \times \frac{2\pi}{60}$$
$$\Omega = \frac{\Delta\theta}{\Delta t} = \frac{\tau}{L} = \frac{Mgr}{I\omega}\ (Precession\ rate\ of\ gyro)$$
$$W = \int \vec{F} \cdot \Delta\vec{x} = |F||\Delta x|Cos(\theta)|$$
$$KE = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$
$$PE = \frac{1}{2}kx_0^2 - \frac{1}{2}kx_f^2$$
$$PE \overset{\text{def}}{=} -W = mgh$$

$$E_{Mech} = \frac{1}{2}Mv_{CM}^2 + \frac{1}{2}I\omega^2 + Mgh_{CM}$$
$$W = \int Frd\theta = \int \tau d\theta = \tau(\theta_f - \theta_i)$$
$$W_{Cons} = -\Delta PE$$
$$\Delta KE + \Delta PE = W_{non-Cons}$$
$$W_{non-Cons} = 0 \Rightarrow \Delta E_{mach} = 0 \Rightarrow E_{mach} = KE + PE = Const$$
Conservative force: $\Delta E_{Mech} = 0$

$$LM1: \vec{F}_{Total} = 0 \Longrightarrow \vec{v} = Const \Longrightarrow \vec{p} = Const$$
$$LM2: \vec{F}_{Total} \neq 0 \Longrightarrow \vec{F} = m\vec{a} \Longrightarrow \vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{L} = \vec{r} \times \vec{p} = rmv = I\vec{\omega}, \frac{d\vec{L}}{dt} = \vec{\tau} = \vec{r} \times \vec{F}$$
$$v_{2A} = \frac{2m_1}{m_1 + m_2}v_{1B} + \frac{m_2 - m_1}{m_1 + m_2}v_{2B}$$

B = Before

$$I = \sum_i m_i r_i^2, Point\ mass: I = MR^2$$

$$Sphere: I = \tfrac{2}{5}MR^2, Cylinder: I = \tfrac{1}{2}MR^2$$
$$Rod\ (about\ end): I = \frac{1}{3}MR^2$$
$$X_{CM} = \frac{1}{M}\int Xdm = \frac{1}{M}\sum_{i=1}^{i=N} m_i X_i$$
$$C = \frac{dQ}{dT}, Q = CdT, c = \frac{C}{m} = \frac{1}{m}\frac{dQ}{dT},$$
$$E = \frac{\sigma}{\epsilon}, p = \frac{F\perp}{A}, \mathrm{K} = -\frac{V\Delta p}{\Delta V}, G = \frac{F \parallel L}{A\Delta x}$$

$$\frac{9}{E} = \frac{3}{G} + \frac{1}{K}, K = \frac{E}{3(1 - 2v)},$$

$$\mu = G = \frac{E}{2(1 + v)}, \lambda = \frac{Ev}{(1 + v)(1 - 2v)}$$
$$\Delta L = \frac{F \perp L}{AE}$$

$$u = \frac{1}{2}E\left(\frac{\Delta L}{L}\right)^2\left(\frac{J}{m^3}\right)(Tensile)$$

$$u = \frac{1}{2}K\left(\frac{\Delta V}{V}\right)^2\left(\frac{J}{m^3}\right)(Bulk\ Stress)$$

$$dF = pA - (p + dp)A - \rho g A dz$$

$$F_{effectiveWeight} = (\rho - \rho_f)gV$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$p + \rho g z + \frac{1}{2}\rho v^2 = const$$

$$\frac{F \parallel}{A} = \frac{\mu \Delta V_x}{L} \Rightarrow \tau = \mu \frac{\partial V_x}{\partial y} \Rightarrow \frac{dp}{dx} = \mu \frac{d^2 v_x}{dz^2}$$

$$\tau(r) = \frac{1}{2}\frac{dp}{dx} r$$

$$Re = \frac{\rho v L}{\mu}, Re = \frac{2R\rho\bar{v}}{\mu}$$

$$x(t) = ACos(\omega_0 t + \phi)$$

$$v(t) = -A\omega_0 Sin(\omega_0 t + \phi)$$

$$a(t) = -A\omega_0^2 Cos(\omega_0 t + \phi) = -\omega_0^2 x(t)$$

$$KE = \frac{1}{2}mv_x^2 = \frac{1}{2}m\omega_0^2 Sin^2(\omega_0 t + \phi)$$

$$PE = \tfrac{1}{2}kx_0^2 Cos^2(\omega_0 t + \phi) = \tfrac{1}{2}kx^2$$

$$\omega = \sqrt{\omega_0^2 - \lambda^2}, \lambda^2 = \frac{b}{2m}$$

$$UnderD: \lambda \ll \omega_0 : x = x_0 e^{-\lambda t} Cos(\omega t + \phi)$$

$$Overdamped: \lambda > \omega_0 : x = Ae^{\mu_+ t} + Be^{\mu_- t}$$

$$Critical\ D: \lambda = \omega_0 : x = (A + Bt)e^{-\frac{b}{2m}t}$$

$$Q < \frac{1}{2} : Od, Q = \frac{1}{2} : CD, Q > \frac{1}{2} : Ud$$

$$v = \lambda f, y = ASin(k(x - vt) + \phi)$$

$$kv = 2\pi f = \omega, v = \sqrt{\frac{\tau}{\mu}}$$

$$\frac{dE}{dx} = A^2\mu\omega^2 Cos^2(kx - \omega t) = \frac{1}{2}A^2\mu\omega^2$$

$$P = vA^2\mu\omega^2 Cos^2(kx - \omega t)$$

$$Same\ f: 2ACos\left(\frac{\delta}{2}\right)\delta\ is\ d\phi$$

$$v = \frac{\omega_1}{\kappa_1} = \frac{\omega_2}{k_2}, \omega = \frac{\omega_1 + \omega_2}{2}, k = \frac{k_1 + k_2}{2}$$

$$Nodes: x = \frac{n\lambda}{2}\ Antinotes: x = \left(n + \frac{1}{2}\right)\frac{\lambda}{2}$$

$$M = \frac{v_{Source}}{v_{sound}}, Sin(\theta) = \frac{v}{v_s}$$

$$\beta = 10\log_{10}\left(\frac{I}{I_0}\right)[dB]\ \ I_0 = 10^{-12}\ [\frac{W}{m^2}]$$

$$p(h) = \int_{z=0}^{z=-h} -\rho g dz = p_0 + \rho g h$$

$$v_t = \frac{(\rho_f - \rho)Vg}{b} = \sqrt{\frac{(\rho_f - \rho)Vg}{c}}$$

$$(p_1 - p_2)dV = \frac{1}{2}\rho dV(v_2^2 - v_1^2) + \rho g dV(z_2 - z_1)$$

$$a_x = -\frac{1}{\rho}\frac{dp}{dx}$$

$$v(x) = -\frac{H^2}{2\mu}\frac{dp}{dx}\left[\frac{H}{z} - \left(\frac{H}{z}\right)^2\right]$$

$$Q = \frac{dp}{dx}\frac{\pi R^4}{8\mu}, \bar{v} = \frac{Q}{A} = \frac{dp}{dx}\frac{R^2}{8\mu}$$

$$A\ddot{x} + B\dot{x} + Cx = 0$$

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{l}} = \sqrt{\frac{mgh}{I}} = \sqrt{\frac{c}{I}} = \frac{2\pi}{T}$$

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{g}{l}} = 2\pi\sqrt{\frac{mgh}{I}} = 2\pi\sqrt{\frac{c}{I}}$$

$$\ddot{\theta} + \frac{g}{L}Sin(\theta) = 0 \Rightarrow \theta = \theta_0 Cos(\omega_0 t + \phi)$$

$$\ddot{\theta} + \frac{c}{I}\theta = 0 \Rightarrow \theta = \theta_0 Cos(\omega_0 t + \phi)$$

$$PE = mg\big(L - LCos(\theta)\big) \approx \frac{1}{2}mgL\theta^2$$

$$E_T = \frac{1}{2}kx_0^2 = \frac{1}{2}kx_0^2\left(e^{-\frac{b}{m}t}\right)$$

$$\zeta = \frac{\lambda}{\omega_0}, A\ddot{x} + B\dot{x} + Cx = 0 \Rightarrow Q = \frac{\sqrt{AC}}{B}$$

$$\mu_\pm = -\frac{b}{2m} \pm \sqrt{\frac{b^2}{4m^2} - \frac{k}{m}}$$

$$A^2 = \frac{F_{Drive}^2}{4m^2\left((\omega_0^2 - \omega_{Drive}^2) + (2\lambda\omega_{drive})^2\right)}$$

$$\kappa = \frac{1}{\lambda} = \frac{k}{2\pi}\ [m^{-1}]\ \ k = \frac{2\pi}{\lambda}\ [\frac{rad}{m}]$$

$$dE = dK + dU = \frac{1}{2}\mu\left(\frac{\partial y}{\partial t}\right)^2 dx + \frac{1}{2}\tau\left(\frac{\partial y}{\partial x}\right)^2 dx$$

$$A^2\mu\omega^2 = \left[\frac{1}{2}\mu\omega^2 A^2 + \frac{1}{2}\tau k^2 A^2\right]$$

$$P_{Av} = \frac{1}{2}vA^2\mu\omega^2$$

$$diff\ f: y = 2ASin(\bar{k}x)Cos\left(\frac{\Delta k}{2}x\right)$$

$$2ACos\left(\frac{\Delta k}{2}\right), f_{beat} = |f_1 - f_2|$$

$$f' = \left(\frac{v \pm v_D}{v \pm v_s}\right)f = f(\frac{v - (v_D - v_m)}{v - (v_s - v_m)})$$

$$f' \approx \left(1 \pm \frac{v_{Rel}}{v}\right)f, In: +ve, out - ve$$